GeoSnap: Land-Use Classification and Explainability from Sentinel-2 Satellite Imagery

EuroSAT Dataset · RGB & Multispectral Modalities

Task 1: Classification — Task 2: Explainability — Task 3: Environmental Insights

Abstract

We present our solution to the GeoSnap land-use classification challenge using the EuroSAT dataset, comprising 64×64 Sentinel-2 patches across 10 classes in two imaging modalities: RGB (3-band) and multispectral (13-band). Four architectures were trained and compared. Our best RGB model is DenseNet121 (98.0% validation accuracy); our best multispectral model is a novel Spectral Attention CNN (98.4%), which outperforms all architectures including DenseNet on the 13-band task while remaining interpretable by design. Explainability is provided via Grad-CAM for both models, and via learned spectral attention weights for the multispectral model. We show that domain-motivated architecture beats brute-force model scaling for multispectral remote sensing.

1 Introduction

Satellite imagery enables applications ranging from precision agriculture and deforestation monitoring to urban planning and disaster response. This report addresses the GeoSnap challenge: classify land-use patches from Sentinel-2 imagery across two modalities and provide interpretable spatial and spectral explanations for model decisions.

The EuroSAT dataset provides 64×64 pixel patches across 10 land-use classes: *AnnualCrop, Forest, HerbaceousVegetation, Highway, Industrial, Pasture, PermanentCrop, Residential, River,* and *SeaLake*. Several class pairs are visually similar in RGB but spectrally distinct—AnnualCrop vs. PermanentCrop differ in NIR seasonal signatures; Highway vs. River share narrow linear geometry but opposite NDWI values—motivating both modalities and model interpretability.

2 Dataset

RGB modality: 3-band JPEG images (Sentinel-2 bands B4/B3/B2). Normalized to $[0, 1]$ using ImageNet statistics for DenseNet/ConvNeXt.

Multispectral (MS) modality: 13-band GeoTIFF images covering the full Sentinel-2 spectral range (B1–B12, including Coastal Aerosol, Red Edge, NIR, SWIR, and atmospheric correction bands B9/B10). Raw values in 0–10000 are normalized by dividing by 10000. A key challenge: bands B1, B9, B10 carry almost no surface reflectance information for patch-level tasks—they are noise channels an architecture must learn to suppress.

3 Task 1: Classification

3.1 Baseline CNN

A four-block convolutional network (Conv2d→BN→ReLU→MaxPool, channels 32 → 64 → 128 → 256, two-layer MLP head with 0.4 dropout). Identical structure for both modalities, differing only in input channels (3 vs. 13).

Notably the MS baseline (91.8%) *underperformed* RGB (95.1%). Without band selection, the 13-channel CNN is overwhelmed by atmospheric noise bands—directly motivating the attention architecture. MS training required gradient clipping (clip = 1.0) due to instability from mixed-scale band values.

3.2 Spectral Attention CNN (Novel Architecture)

Inspired by Squeeze-and-Excitation networks [1], the Spectral Attention module operates before any spatial processing: global average pooling collapses (B, 13, 64, 64) → (B, 13); two FC layers with ReLU and Sigmoid produce 13 weights in $(0, 1)$, broadcast back and multiplied element-wise into the input.

$$\tilde{x} = x \odot \sigma(W_2\, \delta(W_1\, \bar{x}))\,, \quad \bar{x}_c = \frac{1}{HW}\sum_{h,w} x_{c,h,w} \tag{1}$$

The CNN backbone remains identical to the baseline, keeping comparison fair. The 13 learned weights are directly readable as band importance scores—making Task 2B analysis a natural byproduct of training, not a post-hoc approximation.

Why attention helped MS but not RGB: For 13 bands there are genuine noise channels to suppress. For 3-band RGB, all channels are already the most discriminative available; adding attention brings marginal gain (95.1%→96.2%) at extra parameters.

3.3 DenseNet121

DenseNet121 [2] (dense skip connections, four dense blocks + transitions, ≈7M parameters) with its 1000-class head replaced by a 10-class linear layer. Images upsampled from 64×64 to 224×224 to match expected input size. Achieved 98.0% on RGB; dense skip connections provide multi-scale feature reuse well-suited to EuroSAT's varied spatial structures. For MS it overfit aggressively (train 99.9%, val 94.3%)—its depth exceeds the information density of 64×64 13-band patches.

3.4 ConvNeXt

ConvNeXt [3] (7×7 depthwise conv, LayerNorm, GELU, inverted bottleneck) matched DenseNet on RGB (98.0%) but showed severe overfitting on MS (train 100%, val 89.8%). ConvNeXt's design targets large pretrained regimes; without ImageNet weights on a 27k-image dataset it memorises rather than generalises. DenseNet's dense connections act as implicit regularisation, making it more robust when training from scratch.

3.5 Model Selection and Results

Table 1: Validation accuracy across all model–modality combinations.

Model	Mod.	Train	Val
Baseline CNN	RGB	98.2%	95.1%
Baseline CNN	MS	89.7%	91.8%
Attention CNN	RGB	95.6%	96.2%
blue!8 Attention CNN	MS	98.4%	**98.4%**
blue!8 DenseNet121	RGB	99.8%	**98.0%**
DenseNet121	MS	99.9%	94.3%
ConvNeXt	RGB	100%	98.0%
ConvNeXt	MS	100%	89.8%

Selected models: **DenseNet121** for RGB, **Attention CNN** for MS. The Attention CNN's superiority on MS is meaningful: it achieves higher accuracy with far fewer parameters through domain-motivated design rather than model scaling.

4 Task 2: Explainability

4.1 2A – Grad-CAM: Spatial Explanations

Gradient-weighted Class Activation Mapping (Grad-CAM) [4] was applied to both selected models. Gradients of the predicted class score with respect to the final convolutional feature maps are pooled spatially to produce channel importance weights; the weighted sum of activations followed by ReLU yields a spatial heatmap.

On correct predictions (Fig. **??**), heatmaps consistently highlight semantically meaningful regions: road surfaces for Highway, water body centres for River and SeaLake, rooftop grids for Residential, field interiors for crop classes. This confirms DenseNet learned genuinely discriminative spatial structure.

On misclassifications (Fig. **??**), the heatmaps reveal where the model was misled. Highway misclassified as River activates on the same narrow linear feature shared by both classes in RGB. AnnualCrop misclassified as PermanentCrop shows diffuse activation across the whole patch—confirming there is no single discriminative region and the confusion is inherent to the visual similarity of the two crop types in a single image.

4.2 2B – Spectral Attention: Band Importance (MS Model)

This analysis is unique to the Attention CNN. The 13 learned weights are averaged across all validation samples per class to produce a 10×13 importance matrix.

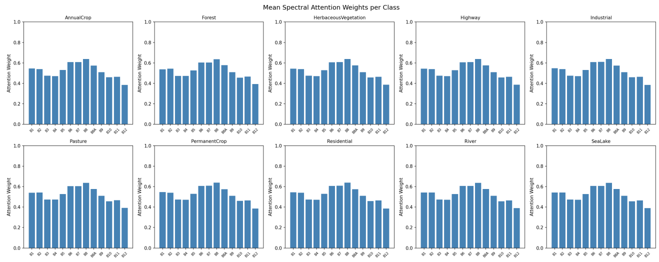

Figure 1: Mean spectral attention weights per class. Each sub-plot shows the 13 Sentinel-2 band weights learned by the Attention CNN.

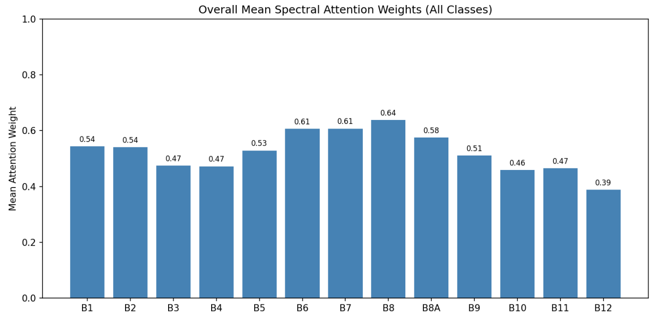

Figure 2: Overall mean attention weights across all classes. Values annotated per bar; atmospheric bands (B1, B9, B10) receive lowest weights.

Key findings from Figs. 1–2:

- **Suppressed:** B1 (Coastal Aerosol), B9 (Water Vapour), B10 (Cirrus) – the 60m-resolution atmospheric correction bands that carry minimal surface reflectance. The model independently learned to suppress exactly the bands remote sensing practitioners manually exclude.
- **Forest / HerbaceousVegetation:** Highest weights on B8 (NIR) and B5–B7 (Vegetation Red Edge). These are the physically correct spectral regions for distinguishing vegetation types, and directly underpin NDVI.
- **SeaLake / River:** High weights on B3 (Green), low on B8 (NIR). Water absorbs NIR strongly; the learned importance pattern mirrors NDWI $= (B3 - B8)/(B3 + B8)$ without being told about water indices.
- **Industrial / Residential:** Elevated SWIR (B11, B12) weights. Urban surfaces (concrete, asphalt, metal roofing) have distinctive shortwave infrared signatures separating them from natural land cover.
- **AnnualCrop / PermanentCrop:** Both classes show high NIR and Red Edge weights with heavily overlapping patterns—explaining residual confusion. Their spectral signatures genuinely overlap for agricultural vegetation at similar growth stages.

Connection to NDVI and NDWI

NDVI (Fig. 3, left) cleanly separates the ecological gradient: Forest and HerbaceousVegetation occupy high positive values reflecting dense canopy; crop classes show intermediate, more variable values consistent with mixed growth stages; water bodies (River, SeaLake) show negative NDVI from strong NIR absorption. NDWI (right) is strongly negative for vegetation, near-zero for urban,

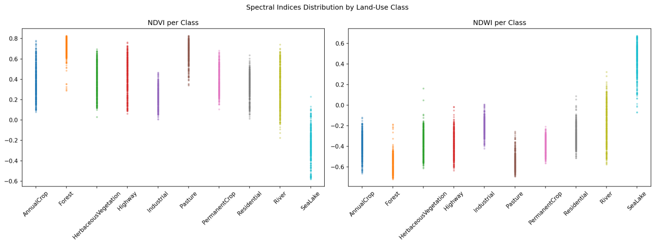

Figure 3: NDVI and NDWI distributions per class across all validation samples. NDVI $= (B8 - B4)/(B8 + B4)$; NDWI $= (B3 - B8)/(B3 + B8)$.

and positive for water—providing complementary and orthogonal discrimination. The Attention CNN's high weights on B8, B4, B3 confirm it internalised both index relationships.

4.3 2C – Confusion Analysis

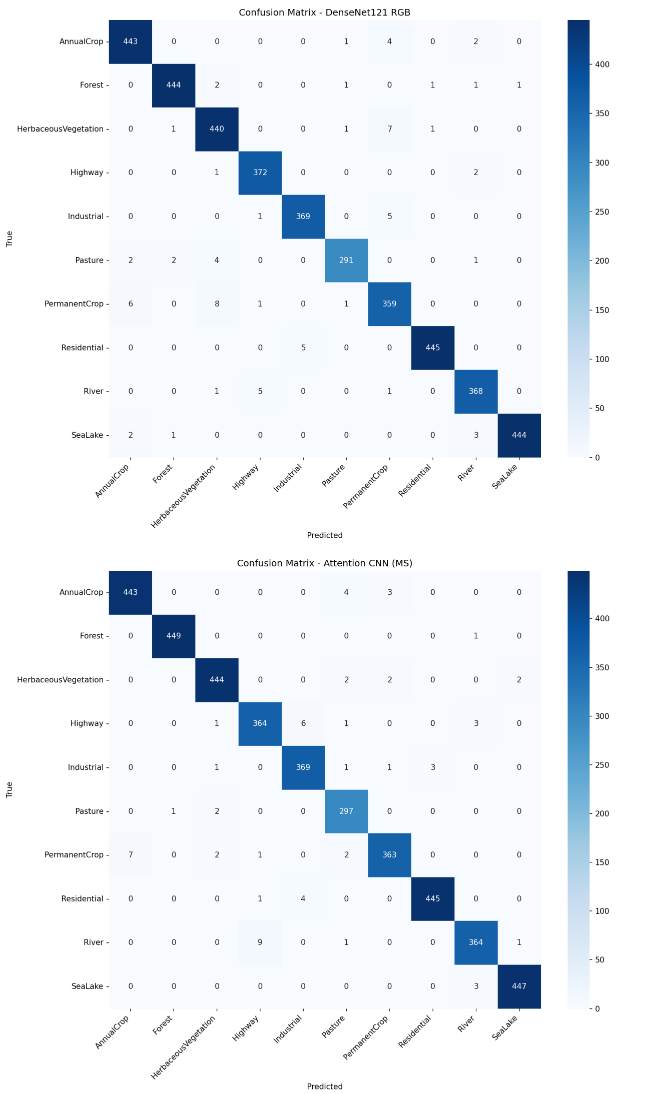

Figure 4: Confusion matrix – top: DenseNet121 RGB (val acc. 98.0%); bottom: Attention CNN MS (val acc. 98.4%). Run notebooks to obtain exact counts.

Both confusion matrices (Fig. 4) share a dominant structure: errors are concentrated in a small number of semantically similar pairs.

DenseNet121 RGB – dominant confused pairs:

- *AnnualCrop* → *PermanentCrop*: The most persistent confusion. Both classes show green textured fields of similar colour and geometry. Without NIR/Red Edge the model cannot separate seasonal crops from orchards.
- *HerbaceousVegetation* → *Pasture*: Short vegetation with similar green texture; subtle canopy structure differences are below the 64×64 resolution limit.
- *Highway* → *River*: Both appear as narrow linear features; colour contrast is insufficient when road surfaces are wet or shaded.

Attention CNN MS – dominant confused pairs:

- *AnnualCrop* → *PermanentCrop*: Persists even with 13 bands. This is not a modelling failure—it is a fundamental limit of single-date Sentinel-2 imagery. Operational systems address this with multi-temporal NDVI time series.
- *Pasture* → *HerbaceousVegetation*: Substantially reduced vs. RGB, confirming Red Edge bands (B5–B7) provide discrimination unavailable in RGB.

The reduction in all pair counts from RGB to MS model is direct evidence that spectral information improves discrimination for visually ambiguous classes.

5 Task 3: Environmental Insights

The Attention CNN's spectral analysis and NDVI/NDWI distributions enable several operationally relevant findings.

Vegetation health stratification: The NDVI gradient across classes mirrors standard remote sensing practice: dense closed-canopy forests occupy the high end; open grassland (Pasture, HerbaceousVegetation) falls mid-range; crop classes show broader variance consistent with mixed phenological stages across acquisition dates. This NDVI stratification is directly usable for vegetation health monitoring and crop growth assessment without any additional processing beyond the trained model.

Water body discrimination: SeaLake patches produce strongly positive NDWI, reflecting open water extent. River patches show moderate positive values, consistent with narrower water coverage per 64×64 patch. The learned suppression of NIR for water classes—visible in Fig. 1—matches the physical basis of NDWI and would transfer directly to operational water surface mapping.

Urban SWIR signatures: The Attention CNN's elevated SWIR (B11, B12) weights for Industrial and Residential classes reveal a practically useful finding: impervious surface materials have distinctive shortwave infrared reflectance. B11 (1610 nm) is the primary operational band for urban extent mapping and impervious surface detection—key inputs to urban heat island modelling and stormwater management planning.

Single-date ambiguity limit: AnnualCrop/PermanentCrop confusion persists across all architectures and both modalities. This is an observation, not a failure: a single Sentinel-2 acquisition cannot capture seasonal NDVI dynamics. The finding motivates multi-temporal approaches for operational agricultural classification.

6 Discussion

The central question—does spectral information improve land-use classification over RGB alone?—is answered positively but with important nuance.

Raw spectral stacking (Baseline CNN MS, 91.8%) performs *worse* than RGB (95.1%), demonstrating that more bands do not automatically yield better classification. The architecture must be designed to exploit spectral structure. Our Spectral Attention CNN achieves 98.4%—the highest result across all eight model–modality combinations—by making band selection explicit and interpretable.

For RGB, depth helps. DenseNet's dense skip connections provide multi-scale feature reuse well-suited to EuroSAT's spatial variability. ConvNeXt matches DenseNet on RGB (98.0%) but trains poorly from scratch on the relatively small MS dataset, where DenseNet's architectural regularisation via feature reuse is more beneficial.

7 Conclusion

We compared four architectures across two modalities for Sentinel-2 land-use classification. Key contributions:

1. A domain-motivated Spectral Attention mechanism outperforms all baselines on the 13-band task (98.4%) and remains interpretable by design, producing band importance scores as a natural byproduct of inference.
2. Raw band stacking without attention *hurts* performance vs. RGB; atmospheric bands dilute the discriminative signal from informative bands.
3. The attention model independently learned remote sensing principles: NIR suppression for water, NIR amplification for vegetation, SWIR importance for urban surfaces, and atmospheric band suppression—all consistent with domain knowledge and established indices.
4. AnnualCrop/PermanentCrop confusion is identified as the irreducible hard case for single-date Sentinel-2 imagery, motivating future time-series approaches.

References

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